

UNIT III – GRAMMARS AND APPLICATIONS – 30 MCQs

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* Indicates required question

MCQ

29. If L is a context-free language, the Pumping Lemma states that sufficiently long strings can be divided into: *

- ☐ Two parts
- ☐ Three parts
- ☐ Four parts
- ☒ Five parts

5. A production of the form $A \rightarrow \alpha$, where A is a single non-terminal, is characteristic of: *

- ☐ Type-0 grammar
- ☐ Type-1 grammar
- ☒ Type-2 grammar
- ☐ Type-3 grammar



16. A production of the form $A \rightarrow \epsilon$ is called: *

- ☐ Unit production
- ☒ Null or epsilon production
- ☐ Terminal production
- ☐ Binary production

27. Which operation preserves Context-Free Languages? *

- ☐ Complement
- ☐ Intersection of two CFLs
- ☒ Concatenation
- ☐ Difference

25. Context-Free Languages are closed under: *

- ☒ Union
- ☐ Intersection
- ☐ Complement
- ☐ Difference



26. Context-Free Languages are NOT generally closed under: *

- ☐ Union
- ☐ Concatenation
- ☐ Kleene star
- ☒ Intersection

19. In Chomsky Normal Form, the production $A \rightarrow BC$ contains: *

- ☐ Two terminals
- ☒ Two non-terminals
- ☐ One terminal and one non-terminal
- ☐ Three non-terminals

2. Which of the following is a component of a grammar? *

- ☒ Production rules
- ☐ Transition function only
- ☐ Stack alphabet only
- ☐ Final states



15. Which of the following is NOT normally a step in simplifying a CFG? *

- ☐ Removing useless symbols
- ☐ Removing ϵ -productions
- ☐ Removing unit productions
- ☒ Adding new DFA states

20. In Chomsky Normal Form, a production of the form $A \rightarrow a$ contains: *

- ☒ One terminal
- ☐ Two terminals
- ☐ One non-terminal
- ☐ Two non-terminals

7. In a CFG, the symbol S generally represents: *

- ☒ Start symbol
- ☐ State symbol
- ☐ Stack symbol
- ☐ Separator



28. The Pumping Lemma for CFLs is primarily used to: *

- ☐ Construct a DFA
- ☒ Prove that certain languages are not context-free
- ☐ Convert CFG to CNF
- ☐ Minimize a PDA

1. A grammar is formally represented as: *

- ☐ $G=(Q,\Sigma,\delta,q_0,F)$
- ☒ $G=(V,T,P,S)$
- ☐ $G=(V,E,\delta,S)$
- ☐ $G=(S,V,T,Q)$

14. The leaf nodes of a parse tree, when read from left to right, give: *

- ☐ The grammar
- ☐ The derivation sequence
- ☒ The generated string
- ☐ The set of non-terminals



4. Type-2 grammars in the Chomsky hierarchy are: *

- ☒ Context-Free Grammars
- ☐ Regular Grammars
- ☐ Context-Sensitive Grammars
- ☐ Unrestricted Grammars

23. In Greibach Normal Form, the first symbol on the right-hand side must be: *

- ☐ A non-terminal
- ☒ A terminal
- ☐ The start symbol
- ☐ An epsilon symbol

17. A production of the form $A \rightarrow B$, where both A and B are non-terminals, is called: *

- ☐ Null production
- ☒ Unit production
- ☐ Terminal production
- ☐ Binary production



24. Which of the following is an application of Context-Free Grammars? *

- ☒ Syntax specification of programming languages
- ☐ IP address assignment only
- ☐ Hardware manufacturing
- ☐ Image compression only

12. A parse tree is also called a: *

- ☐ Transition tree
- ☒ Derivation tree
- ☐ State tree
- ☐ Finite tree

10. In a leftmost derivation, which symbol is replaced first? *

- ☐ Rightmost terminal
- ☒ Leftmost non-terminal
- ☐ Rightmost non-terminal
- ☐ Leftmost terminal



3. According to the Chomsky hierarchy, Type-0 grammars are called: *

- ☐ Context-Free Grammars
- ☐ Regular Grammars
- ☒ Unrestricted Grammars
- ☐ Context-Sensitive Grammars

18. Which normal form requires productions mainly of the form $A \rightarrow BC$ or $A \rightarrow a$? *

- ☐ Greibach Normal Form
- ☒ Chomsky Normal Form
- ☐ Regular Normal Form
- ☐ Reduced Normal Form

8. Which of the following is a valid CFG production? *

- ☐ $AB \rightarrow a$
- ☒ $A \rightarrow aB$
- ☐ $ab \rightarrow AB$
- ☐ $aB \rightarrow bA$



11. In a rightmost derivation, which symbol is replaced first? *

- ☐ Leftmost non-terminal
- ☐ Leftmost terminal
- ☒ Rightmost non-terminal
- ☐ Start symbol only

22. In Greibach Normal Form, productions generally have the form: *

- ☐ $A \rightarrow BC$
- ☒ $A \rightarrow a\alpha$
- ☐ $A \rightarrow \epsilon$ only
- ☐ $A \rightarrow AB$

6. Which machine recognizes Context-Free Languages? *

- ☐ Finite Automaton
- ☒ Pushdown Automaton
- ☐ Turing Machine only
- ☐ Linear Bounded Automaton



21. In Chomsky Normal Form, the start symbol may produce: *

- ☐ Only two terminals
- ☒ ϵ , under the usual exception when ϵ belongs to the language
- ☐ Any arbitrary string
- ☐ Only another start symbol

9. A derivation in a grammar is a sequence of: *

- ☐ State transitions
- ☒ Rewriting steps using production rules
- ☐ Stack operations
- ☐ DFA transitions

13. In a parse tree, the root node represents: *

- ☐ A terminal symbol
- ☐ The first production
- ☒ The start symbol
- ☐ The last terminal



30. In the Pumping Lemma for CFLs, a string z is typically represented as: *

- ☒ $z=uvwxy$
- ☐ $z=uvwx$
- ☐ $z=abcde$ only
- ☐ $z=xyz$

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